

Process Characterization Report

A-Mab Drug Substance — Ultrafiltration / Diafiltration (Step 10) — PCR-010

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—

Role	Function	Name (synthetic)	Signature	Date
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AMV, analytical method validation; Cpk, process capability index; CPP, critical process parameter; CQA, critical quality attribute; DS, drug substance; DV, diavolume; HCP, host cell protein; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; RPN, risk priority number; RSD, relative standard deviation; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; TFF, tangential flow filtration; TMP, transmembrane pressure; UF/DF, ultrafiltration and diafiltration; UV, ultraviolet absorbance at 280 nm; WC-CPP, well controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Ultrafiltration and diafiltration is the last unit operation of the A-Mab drug substance process, and it concentrates the virus filtrate, exchanges it into the formulation buffer and delivers the drug substance at its target concentration. The step forms no critical quality attribute and is credited with no impurity or virus clearance. The studies reported here were executed against PCP-010 on a qualified scale-down model of the commercial tangential flow filtration skid (SOP-1001 and SOP-2013), and their scope was set by the pre-characterization risk assessment RA-001. This report is the Stage 1 process design record for the step in the sense used by the process validation lifecycle guidance (U.S. Food and Drug Administration 2011), and the development approach behind it is the enhanced approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012).

The characterization covered 3 parameters, which are the number of diavolumes, the transmembrane pressure and the final drug substance concentration, and all of them are classified as key process parameters. None is linked to a critical quality attribute, so the step contributes no element to the design space and no proven acceptable range is claimed for any of its parameters. At its set-points the step delivered a product yield of 97.1% and 54.6 kg of drug substance at 75 g/L, which closes the train at a cumulative product yield of 83.2%.

The drug substance attributes measured on the pool this step delivers meet their acceptance criteria at commercial scale. Across 2,000 simulated batches, with every process parameter of every step drawn about its set-point at one sixth of its normal operating range, aggregate reached at most 1.15 % HMW against a limit of 5 %, and acidic charge variants reached at most 27.2 % against a limit of 40 %. Acidic charge variants carry the tightest capability of the three attributes confirmed here, at a Cpk of 4.26. The tightest capability of the drug substance as a whole belongs to Viral clearance — MVM (parvovirus), at a Cpk of 1.51, and it is credited to anion exchange and small virus retentive filtration (PCR-008 and PCR-009), which are the two steps that clear parvovirus.

Execution recorded 2 deviations, both of which were investigated and impact assessed, both of which were dispositioned as retained, and neither of which changed a parameter range, a classification or a conclusion of this report (§13). The step delivers the drug substance at its target concentration and, by the diavolume count set in §4.1, in the formulation buffer, within the ranges given in Table 4.1 and on the scale-down model described in §3.1. Residual load buffer was not assayed, and the completeness of the exchange is taken from the washout relation given in §4.1. The step assures no critical quality attribute on its own, and its outcome rolls up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose principal mechanism of action is antibody dependent cellular cytotoxicity. The drug substance process runs from the production bioreactor through harvest and clarification, Protein A capture, low pH viral inactivation, cation exchange, anion exchange and small virus retentive filtration to this step, which is the formulation operation of that train. It takes the virus filtrate, brings it to the drug substance concentration and buffer, and hands the material on for storage and fill.

The operation is a tangential flow filtration process, in which the retentate is recirculated across a membrane whose nominal cut off retains the antibody while the permeate carries the small solutes (buffer salts, excipients and residual process chemicals) to waste. It runs in three phases. The load is first concentrated, then diafiltered against the formulation buffer for a set number of diavolumes, then concentrated a second time to the target and recovered from the skid with a buffer flush.

Each of the three parameters acts through a different physical mechanism. The number of diavolumes sets how completely the load buffer is replaced by the formulation buffer, while transmembrane pressure sets the permeate flux and with it both the processing time and the shear the product sees at the membrane surface. The final concentration sets the concentration of the drug substance and the viscosity at which the pool is recovered. The product leaves the step at 75 g/L, which is about 17 times the titre of the harvested culture fluid.

No critical quality attribute is formed at this step and none is cleared. The glycan, charge variant and aggregate attributes are established in the production bioreactor (PCR-003), and host cell protein, residual DNA and leached Protein A are cleared by the chromatography steps (PCR-005, PCR-007 and PCR-008). Viral safety rests on low pH inactivation, anion exchange and small virus retentive filtration (PCR-006, PCR-008 and PCR-009). Some clearance of small solutes occurs across any membrane operation, but it is not predictable and no claim is made for it, so the impurity levels entering this step are taken to carry through unchanged to the drug substance.

1.2 Objectives

The study was designed to meet the objectives listed below.

- (i) Confirm that the step delivers drug substance at its target concentration and closes the product mass balance for the batch.
- (ii) Establish a normal operating range and a wider characterization range for each of the parameters that govern the operation.
- (iii) Confirm that the drug substance quality attributes measured after the step remain within their acceptance criteria.
- (iv) Classify each parameter under the decision logic of SOP-4001, and record the evidence the classification rests on.
- (v) Provide the ranges and the controls that the Stage 2 process performance qualification will be run against.

1.3 Regulatory and scientific basis

The characterization follows the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), and this report is the Stage 1 record for the step, so the ranges it establishes are the ranges Stage 2 qualifies. Development follows the enhanced approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012), in which process understanding and quality risk management together decide what is controlled and how tightly (International Council for Harmonisation 2023b). Criticality is treated as a continuum and not as a binary label, which is the convention used throughout the A-Mab case study (CMC Biotech Working Group 2009).

ICH Q5A governs the evaluation of viral safety (International Council for Harmonisation 2023a), and it is named here only to record that this step contributes nothing to the viral clearance claim for the drug substance. The modular clearance argument is built in PCR-006, PCR-008 and PCR-009 and is consolidated in PCMR-001.

1.4 Related documents

The documents this report depends on, and the document it rolls up into, are listed in Table 1.1. The paired protocol is PCP-010, which fixed the study design and the acceptance criteria before any data existed, while the controlled procedures and the validated analytical methods used at the step are listed separately in §3.3.

Table 1.1: Related documents.

Docu- ment ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope

Docu- ment ID	Title	Relationship
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-010	Process Characterization Plan (Ultrafiltration / Diafiltration (Step 10))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior product knowledge

Formulation by tangential flow filtration is a platform operation at Novacyte Biologics, and the same membrane chemistry, the same formulation buffer and the same membrane loading have been used for three earlier humanized IgG1 products (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical supply. That history is why the step is characterized against a known mechanism instead of developed from first principles. The mechanism is also simple enough to state in advance, since a membrane that retains the antibody and passes small solutes changes the composition of the pool and leaves the product alone. The expected outcome of the study was therefore an absence of effects on product quality.

Two mechanisms could break that expectation, and both were carried into the study. Shear at the membrane surface can generate aggregate, and transmembrane pressure is the parameter that governs it. Holding the antibody at high concentration through the second concentration phase can do the same, and the final concentration is the parameter that governs that. RA-001 scored both as low risk on the platform history recorded there, but they are nevertheless the reason the characterization ranges were set wider than the normal operating ranges.

2.2 Quality attributes in scope

Three drug substance attributes are measured on the pool this step delivers, and they are the attributes in scope for this report (Table 2.1). None of them is set at this step.

Table 2.1: Drug substance quality attributes measured after the step.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18-40 % (CEX/icIEF)	VL	4
Residual DNA	process impurity	0-0.001 ng/dose	VL	6

Aggregate is the attribute with a mechanism at this step. It is formed in the production bioreactor (PCR-003) and reduced principally at cation exchange (PCR-007), and the concentration and recirculation applied here are the last opportunity in the process to add to it. Acidic charge variants are formed in the bioreactor and are challenged by the low pH hold (PCR-006), and they are measured again here because the drug substance is the material the specification applies to. Residual DNA is cleared across the three chromatography steps and is reported for the drug substance on a per dose basis, and that conversion of basis is made at this step.

The remaining attributes of the register are not repeated here. The glycan attributes, host cell protein and leached Protein A are covered by the reports for the steps that set or clear them, and the full register with those attributions is given in Appendix C.

2.3 Risk based prioritization and parameter selection

RA-001 carried 3 parameters into the pre-characterization assessment for this step and assigned all of them to univariate study. The scoring is reproduced in Table 2.2.

Table 2.2: Pre-characterization risk ranking and study assignment for the step (RA-001).

Parameter	Prospective failure mode	Effect	Sev.	Occ.	Det.	Initial RPN	Assigned study	Priority
Number of diavolumes	Diavolumes < 7	Incomplete buffer exchange (formulation; not part of DS characterization)	4	4	6	96	univariate	Low
Transmembrane pressure	TMP outside 10-15 psi	Affects flux/processing (formulation)	4	4	6	96	univariate	Low
Final DS concentration	Final concentration outside 65-85 g/L	DS concentration spec (formulation)	4	4	6	96	univariate	Low

The three parameters scored alike, and the reason is visible in the effect column, where none of the failure modes reaches a quality attribute. Severity was scored in the Slight band, which the A-Mab scale uses for an unlikely impact on product quality that calls for no supplemental testing and no reprocessing. Occurrence was scored in the Low band and detection in the Moderate band, and the detection band is the one the scale uses for a failure that is seen at a later operation and not during this one.

The three scores together give an initial risk priority number of 96 for each parameter. On that basis the assessment concluded that a designed experiment was not justified and that univariate study of each range would support the operating limits the step needs.

A low score is not the same as no study. The parameters still set the concentration and the buffer the drug substance is delivered in, and both of those are specified, so what the score decided was the kind of study and not whether one was needed.

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were executed on a scale-down tangential flow filtration system qualified as a model of the commercial skid under SOP-1001. The model holds the scale independent variables of the operation at their commercial values. Those variables are the membrane chemistry and nominal cut off, the membrane loading in product mass per unit area, the crossflow per unit area, the transmembrane pressure, the number of diavolumes and the final concentration. The scale dependent features were set to match commercial performance instead of commercial geometry, which for this operation means the recirculation loop, the pump and the system hold-up volume.

Qualification compared the model with commercial equivalent runs on the quantities that leave the step, which are step yield, final concentration, aggregate and charge variants. The qualification record is held under SOP-1001 and is not reproduced here. The skid used for the studies (EQ-TFF-142) was in calibration at execution, and its next calibration fell due on 2026-04-22.

One feature of the model cannot be matched to the commercial skid by scaling, and it is the hold-up volume of the recirculation loop and the membrane housing, because buffer held there is recovered with the pool at the flush and dilutes it. The effect is managed by the recovery flush described in §3.2, and it is the mechanism behind DEV-010-02 (§13.2).

The warrant this model has to carry is narrow. It supports operating ranges for parameters that govern process performance, and it supports the statement that the drug substance attributes are unchanged across the step, but it is not asked to support a quality attribute claim of its own, because the step makes none.

3.2 Operation

The step was operated as described in SOP-2013, taking as its load the pool from small virus retentive filtration. That pool is concentrated, diafiltered against the formulation buffer for 8 diavolumes, then concentrated to 75 g/L and recovered from the skid with a buffer flush. Transmembrane pressure is held at 12 psi through both concentration phases and through the diafiltration.

Inputs to the step are controlled to platform specification and were not characterized, and they are the membrane, the formulation buffer and the load itself, whose quality is the subject of PCR-009. The membrane is installed, flushed and integrity tested

before use under SOP-2013. The system is cleaned and sanitized between batches under the same procedure.

3.3 Analytical methods

The controlled procedures and the validated analytical methods used at this step are listed in Table 3.1.

Table 3.1: Controlled procedures and validated analytical methods for the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

Protein concentration is measured by ultraviolet absorbance under AMV-3019, and it is both the quantity that defines the endpoint of the second concentration phase and a drug substance release test. Aggregate is measured by size exclusion chromatography under AMV-3011 and charge variants by imaged capillary isoelectric focusing under AMV-3013, and both of these are release tests applied to the final pool of the step. The values this report gives for aggregate and for charge variants are the commercial scale estimates of §8, and no result from an individual run is reported for either. Validated performance for the concentration method is given in Appendix A. Performance for the other two methods is held in their own validation reports.

3.4 Sampling plan

Samples were taken at three points of the operation, which were the load before the first concentration phase, the retentate at the end of the diafiltration, and the final pool after the recovery flush. The final pool is the drug substance, and it is the sample the release panel is applied to. Transmembrane pressure was recorded continuously by the skid through both concentration phases and the diafiltration, which is how the excursion in §13.1 was detected. Retained samples of the load and of the final pool were kept. No result from a retained sample is reported here.

3.5 Statistical methods

No designed experiment was run at this step, and none was planned, so the screening and response surface framework used at the six steps that carry designed experiments does not apply here. A screening design identifies which factors are active on a quality attribute, and a response surface model is the predictive model from which a design space is then read. Both of them need a quality attribute to model. This step governs none, and the analysis is therefore descriptive.

Two quantitative sources support the statements made in this report. The first is the seeded nominal batch, run with every parameter of every step at its set-point, which gives the step performance in Table 5.1. The second is a Monte-Carlo simulation of 2,000 commercial scale batches in which every parameter of every step is drawn independently from a normal distribution centred on its set-point, with a standard deviation of one sixth of its normal operating range, and clipped to its characterization range. The drug substance attributes from that simulation give the capability reported in §8.

Capability is reported as a one-sided index against the upper acceptance limit of each of the three attributes, so Cpk is the distance from the simulated mean to that limit expressed in units of three standard deviations. Aggregate and residual DNA are impurities, and an upper limit is the whole of their acceptance criterion. Acidic charge variants are shown in Table 2.1 with a lower figure as well, and that figure is not the limit capability is taken against here, because the risk the attribute carries at the drug substance is an elevated level. No acceptance threshold for Cpk is adopted in this report, and the values are reported against the acceptance limits themselves, which are the criteria the drug substance is released against.

4 Study design

4.1 Parameters, ranges and the knowledge space

The three parameters, their set-points, their normal operating ranges and their characterization ranges are given in Table 4.1.

Table 4.1: Parameters characterized at the step, with the final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Number of diavolumes	DV	8	7-9	7-10	KPP	univariate
Transmembrane pressure	psi	12	10-14	10-15	KPP	univariate
Final DS concentration	g/L	75	70-80	65-85	KPP	univariate

Each range was set from the mechanism the parameter acts through. Diafiltration removes a fully permeable solute exponentially, so the residual fraction of the load buffer after N diavolumes is e raised to the power minus N . At the set-point of 8 diavolumes that residual is 0.034% of the load concentration. At the low edge of the characterization range it is 0.091% and at the high edge 0.005%, which is a span of about 20-fold in carry-over. The range covers a wide swing in the completeness of the exchange while leaving the exchanged species at trace level throughout it. This is a design calculation for an ideal, fully permeable solute in a well mixed retentate, and it is the basis of the range and not a measured result.

Transmembrane pressure was ranged asymmetrically about its set-point, over 10-15 psi against a normal operating range of 10-14 psi. The mechanism that moves transmembrane pressure during a batch is membrane fouling, and fouling moves it upward, so the characterization range was extended on the high side only. The excursion recorded in §13.1 reached 14.6 psi and therefore fell inside characterized territory.

The final concentration was ranged 10 g/L either side of the target of 75 g/L, with a normal operating range half that wide, and the width of the normal operating range is set by the measurement that controls the endpoint. Intermediate precision of the concentration method is 1.2 %RSD (AMV-3019), which at the target is 0.9 g/L, so the normal operating range is about 5.6 assay standard deviations wide on each side. A range that is narrow relative to the assay controlling it cannot be held, and this range is not narrow on that measure.

4.2 Univariate assessment

The characterization schedule took each parameter to both edges of its characterization range with the other two held at their set-points, and the set-point condition was the reference, with the levels given in Table 4.2.

Table 4.2: Univariate evaluation levels for the three parameters.

Parameter	Unit	Low level	Reference (set-point)	High level	NOR
Number of diavolumes	DV	7	8	10	7-9
Transmembrane pressure	psi	10	12	15	10-14
Final DS concentration	g/L	65	75	85	70-80

Each level was judged on process performance and not on product quality, and the criteria are listed in §5.2. Product quality was assessed on the pool of the set-point batch and on the commercial scale simulation described in §3.5, and not at the edges of the ranges, which is a boundary §5.4 and §7 carry forward.

A multivariate design was neither run nor needed. Interactions matter when two parameters act on the same quality attribute, and no parameter of this step is linked to one in the register, so there is no governed attribute on which an interaction could be estimated or classified. The two parameters that carry a mechanism at all, which are transmembrane pressure through shear and the final concentration through the concentrated hold (§2.1), were both scored low risk in RA-001 on the platform history recorded there. The three parameters also act through different mechanisms and are read out on different responses, and that is the second reason a factorial design would have added cost without adding process understanding. Diavolumes are read out on the composition of the pool, transmembrane pressure on flux and processing time, and the final concentration on the endpoint of the second concentration phase.

The absence of a quality linkage is a property of the attribute register and not a gap in the study, and it is what keeps all three parameters out of the design space and inside the control strategy, where they are held because monitoring them is what keeps the process consistent (CMC Biotech Working Group 2009).

5 Results

5.1 Nominal batch performance and mass balance

The step delivered drug substance at its target concentration and closed the product mass balance for the batch. The nominal batch is given in Table 5.1.

Table 5.1: Nominal batch performance of the step at its set-points.

Metric	Nominal batch
Step yield (fraction of input product mass)	0.971
Product mass out (g)	54,588
Drug substance concentration (g/L)	75
Residual DNA (ng/dose)	0.0001

Product loss across the step was 2.9% of the mass entering it, and the platform attributes it to hold-up volume and to retention on the membrane. Hold-up volume is the mechanism behind DEV-010-02 (§13.2), and membrane retention was not measured separately. The step delivered 54.6 kg of drug substance at 75 g/L, which is 728 L of pool. Cumulative product yield for the train closes at 83.2%, and the step by step figures are given in Appendix B. Residual DNA is reported for the drug substance at 0.0001 ng/dose against a limit of 0.001 ng/dose.

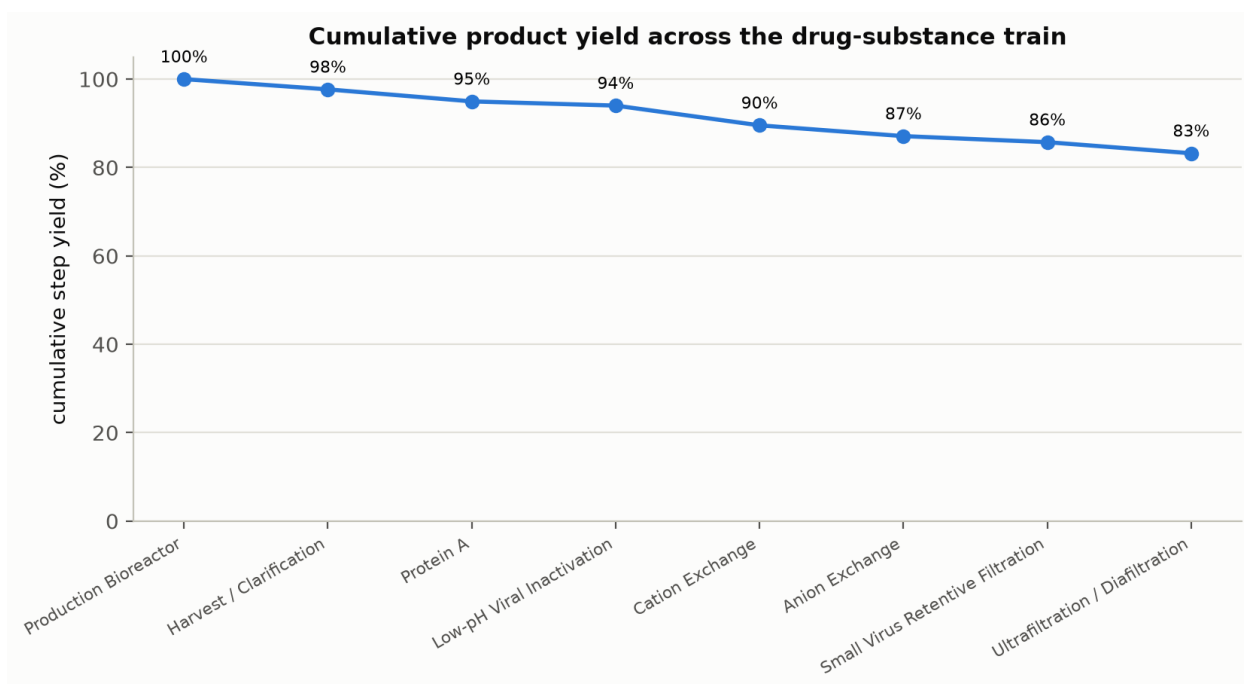


Figure 5.1: Cumulative product yield across the drug substance train, from the production bioreactor to the formulation step.

The shape of Figure 5.1 is the reason the step is reported at all, because it costs product without changing quality, and the loss it costs is of the same order as the losses at the chromatography steps. It therefore belongs in the mass balance argument even though it belongs in no quality argument.

5.2 Characterization levels and process performance

The univariate levels of Table 4.2 were assessed against process performance criteria only, and no product quality result is reported at any of them. The criteria were the permeate flux and the processing time the level produced, the concentration the second phase reached at its endpoint, and closure of the product mass balance at recovery. None of those is a quality criterion, and none of them could be, because no attribute of the drug substance register is set or cleared at this step.

The quantitative basis of this report is the two sources named in §3.5. The set-point batch gives the performance of the step, and the commercial scale simulation gives the quality of the pool it delivers. What the levels add is the width of the region the step was operated over, which is the region the characterization ranges in Table 4.1 describe, and the limitation that leaves is stated in §5.4 and carried into the classification in §9.

5.3 Drug substance concentration and its measurement

The final concentration is the only parameter of this step that is measured on the product itself, which makes it both the easiest to demonstrate control of and the easiest to correct. Intermediate precision of the method is 1.2 %RSD, with a limit of quantitation of 0.5 g/L and accuracy of 99.6 % (AMV-3019, Appendix A). The method accounts for 22.0% of the observed variance in the reported concentration, so most of the spread seen between batches is process and not assay.

One run finished its second concentration phase below target, at 71.5 g/L, and the shortfall of 3.5 g/L is 3.9 times the intermediate precision of the assay at the target, so it is a real difference and not a measurement artefact. The pool was corrected in process, and the investigation is reported in §13.2.

5.4 Product quality across the step

The drug substance attributes measured after the step meet their acceptance criteria, and the quantitative outcome is given in §8. Aggregate (SEC-HPLC, AMV-3011) is the attribute with a mechanism at this step, because the product is concentrated and recirculated across a membrane. Charge variants and residual DNA have no mechanism here at all, and they are measured on the pool because the drug substance is the material their specifications apply to.

The quality confirmation covers the normal operating ranges of the three parameters, which is the region the simulated batches concentrate in. It does not extend to the edges of the characterization ranges, where the evidence from this study is process performance only, and that boundary is carried into the range statements in §6 and §7.

6 Design space

This step contributes no element to the design space of the A-Mab drug substance process. A design space is the multivariate combination of parameters that has been shown to assure quality (International Council for Harmonisation 2009), and the assurance has to be assurance of something. None of the three parameters characterized here is linked to a critical quality attribute, so there is nothing at this step for a design space to assure.

What the step has instead is an operating range for each parameter. The normal operating ranges in Table 4.1 sit inside the wider characterization ranges, and the characterization ranges are the knowledge space for the step, so operation anywhere inside a normal operating range is normal operation. Operation outside a characterization range is a change, and it is assessed under the pharmaceutical quality system before it is made (International Council for Harmonisation 2008).

The corner of the characterized region that challenges the step hardest is the combination of the low diavolume edge, the high transmembrane pressure edge and the high concentration edge. It gives the least complete buffer exchange, the highest shear at the membrane and the highest product concentration at the same time. That corner is bounded by the range edges in Table 4.1, and it is a process performance worst case and not a quality worst case, because no quality attribute is at stake at this step.

These statements are bounded in three ways. They hold over the ranges in Table 4.1 and not beyond them. They rest on the scale-down model described in §3.1, whose warrant is process performance and unchanged quality and not a quality claim of its own. They assume the membrane chemistry, the membrane loading and the formulation buffer used in the study, none of which was varied.

7 Proven acceptable ranges

No proven acceptable range is claimed for any parameter of this step. A proven acceptable range is the range of a parameter over which a named quality attribute stays inside its acceptance criterion, where the range is proven by supporting data, and that definition needs a governed attribute, which this step does not have.

What the study supports instead is the operating range in Table 4.1, and its acceptance basis is process performance. The criteria the ranges were judged against are the target concentration of the drug substance and closure of the product mass balance, and both of those are process criteria and not quality criteria. Buffer exchange was not judged against a measured criterion at all, because it is controlled by the diavolume count, and the residual load buffer follows from the washout relation given in §4.1.

The proven acceptable ranges that bound the attributes measured after this step are established elsewhere. Aggregate is bounded at the production bioreactor and at cation exchange (PCR-003 and PCR-007), charge variants are bounded at the bioreactor and at the low pH hold (PCR-003 and PCR-006), and residual DNA is bounded across the three chromatography steps (PCR-005, PCR-007 and PCR-008). Those reports carry the analysis, and this one carries the confirmation that the drug substance still meets the criteria after the last operation of the process.

That confirmation is bounded to the ranges characterized here and to the attributes listed in Table 2.1, and it says nothing about the attributes this step does not measure, which are covered by the reports named in Appendix C.

8 Process capability and robustness

The drug substance attributes measured after this step meet their acceptance criteria at commercial scale with wide margin, and capability against the upper limit of each is given in Table 8.1.

Table 8.1: Commercial-scale capability of the attributes measured after the step.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Aggregates (HMW)	H	upper	0-5	0.753 $\pm$ 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18-40	22.3 $\pm$ 1.4	4.26
Residual DNA	VL	upper	0-0.001	0.0001 $\pm$ 0	10.8

Aggregate is the most capable of the three, at a Cpk of 16.1. Its mean across the simulated batches is 0.75 % HMW against a limit of 5 %, a margin of 4.25 %, and the highest simulated batch reached 1.15 %. Acidic charge variants carry the tightest capability of the three, at a Cpk of 4.26, with a mean of 22.3 % against a limit of 40 %, a margin of 17.7 %, and a highest simulated batch of 27.2 %. Residual DNA reached at most 0.0003 ng/dose against a limit of 0.001 ng/dose, at a Cpk of 10.8. Its spread across the simulated batches is smaller than the rounding Table 8.1 carries, so the deviation column of that row reads zero while its Cpk is finite.

Commercial-scale process capability (n = 2000 simulated batches)

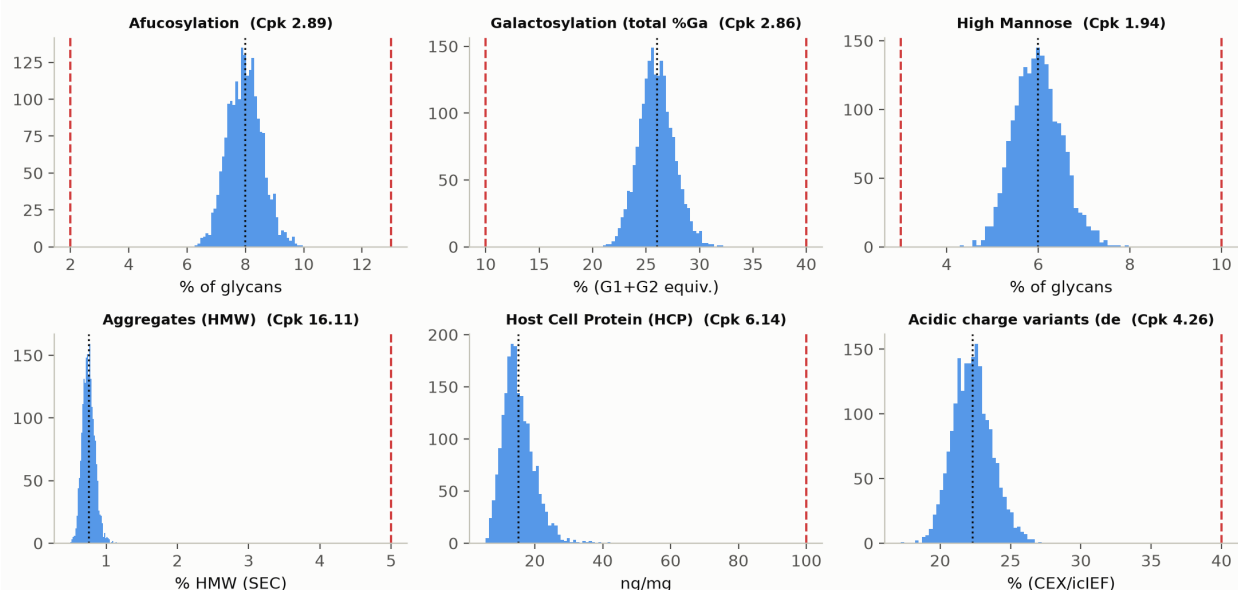


Figure 8.1: Commercial-scale distributions of the glycan attributes, aggregate, host cell protein and acidic charge variants from the Monte-Carlo simulation, with acceptance limits and the capability of each panel.

The simulated distributions are shown in Figure 8.1, which plots the glycan attributes, aggregate, host cell protein and acidic charge variants. Aggregate and acidic charge variants are the two panels in scope for this report. Residual DNA is not among the panels, and its capability is given in Table 8.1. The remaining panels are attributes set or cleared upstream, and they are covered by the reports named in Appendix C.

The simulation varies every parameter of every step about its set-point, including the three parameters of this step, on the sampling described in §3.5. The figures therefore already carry the operating variability this report is responsible for, but they do not isolate it. A step with no effect on an attribute contributes nothing to the spread of that attribute, which is why the capability of the drug substance is not by itself a sensitive test of this step.

The tightest capability of the drug substance as a whole is not one of these three. It belongs to Viral clearance — MVM (parvovirus), at a Cpk of 1.51, and it is credited to anion exchange and small virus retentive filtration (PCR-008 and PCR-009), which are the two steps that clear parvovirus. The low pH hold carries the enveloped virus claim (PCR-006) and contributes nothing to this one. Capability across every attribute of the drug substance is consolidated in PCMR-001.

Two bounds apply to all of these figures. They are estimates from qualified scale-down models under the sampling described in §3.5, and they are not measurements at commercial scale. They carry the batch to batch variation the model reproduces and not the additional variation a commercial facility contributes. Capturing the rest is what Stage 2 is for.

9 Parameter classification

All 3 parameters of this step are classified as key process parameters under the decision logic of SOP-4001, and none is a critical process parameter or a well controlled critical process parameter. The classification of each and the reason for it are given below.

- Number of diavolumes, KPP. It sets the completeness of the buffer exchange, which is a formulation property of the drug substance and not a critical quality attribute, and it is counted from permeate mass and easy to hold. The characterization range spans a wide swing in carry-over, and no attribute of the drug substance register is set or cleared by the step, so no quality criterion bounds the range (§2.2 and §5.4).
- Transmembrane pressure, KPP. It governs the permeate flux and the processing time, and through shear at the membrane it has the most plausible route of the three to a quality effect. It is nevertheless classified on process performance, because this step governs no attribute against which such an effect could be classified. It is monitored continuously, and the one excursion recorded stayed inside the characterized range (§13.1).
- Final DS concentration, KPP. It is the endpoint of the step and it is measured on the drug substance at release, so it is controlled closely for reasons of process consistency and not of quality risk. A low result is correctable in process by further concentration, which is what happened in DEV-010-02.

Across the whole train 10 parameters are classified as key process parameters, of which 3 are at this step. The classification follows from the quality attribute register, in which no drug substance attribute is set or cleared at this step, and not from a demonstrated null effect on one. All three parameters stay in the control strategy because monitoring and controlling them is what keeps the process consistent and predictable (CMC Biotech Working Group 2009).

10 Contribution to the control strategy

The step contributes to the control strategy in four ways, and each of them is a process performance control rather than a quality control.

The first is parameter control, in which the three parameters are held to the normal operating ranges in Table 4.1, and those are the ranges Stage 2 will be run against.

The second is in-process monitoring. Transmembrane pressure is trended on the commercial skid through both concentration phases and the diafiltration, the diavolume count is taken from permeate mass, and the concentration is measured before the pool is released to storage.

The third is release testing on the drug substance, in which concentration, aggregate and charge variants are measured on the final pool under AMV-3019, AMV-3011 and AMV-3013, and those results are the last quality data generated on the batch before it leaves drug substance manufacture.

The fourth is control of the inputs. The membrane and the formulation buffer are controlled to platform specification, and a change to either is assessed before it is made, because neither was varied in this study.

The step also has clear limits. It controls no critical quality attribute, and it provides no impurity clearance credit and no viral clearance credit. Every attribute in Table 2.1 is delivered to it inside its limit by the steps upstream, and the contribution this step makes is to deliver that material at the right concentration in the right buffer without changing it. The consolidated control strategy across the train is in PCMR-001.

11 Discussion

The step is understood well enough to transfer, and part of the reason is that there is little to understand, since a membrane that retains the antibody and passes small solutes changes the buffer and leaves the product alone. The study confirmed that expectation against the drug substance attributes measured on the pool the step delivers, and it established ranges for the three parameters that govern how the step performs. Nothing in the result was unexpected, which is the outcome a platform operation should produce.

Confidence in the scale-down model is high for process performance and adequate for what is claimed about quality. The model matches the commercial skid on the variables that drive the operation, and hold-up volume is the one feature it cannot match by scaling. That feature showed itself during execution as a concentration shortfall on one run (§13.2), and it is managed by the recovery flush and not by a parameter range.

Two deviations were recorded, and the more significant of them is the transmembrane pressure excursion, in which pressure reached 14.6 psi against an upper normal operating limit of 14 psi. It stayed inside the characterization range, the parameter governs no quality attribute, and the run was retained. The second deviation was the concentration shortfall, which was corrected in process, and neither event changed a range, a classification or a conclusion.

Three limitations bound what this report establishes. The first is the absence of a designed experiment, which means that interactions between the three parameters are not quantified and that this study cannot exclude one. The justification for accepting that is the justification for the univariate design itself, which is that no parameter of the step is linked to a quality attribute in the register (§4.2). If a future process change gives one of them such a link, the study type has to be revisited.

The second limitation is that the capability figures in §8 are model estimates and not measurements at scale, and they are confirmed during Stage 2 on batches made in the commercial facility with the commercial skid. The third is that the ranges hold for the membrane chemistry, the membrane loading and the formulation buffer used here. A change to any of them falls outside the knowledge space of this study, and the platform history that supports the step does not extend to a membrane the platform has not been run on.

No parameter of this step needed the most stringent form of control. That conclusion follows from the absence of a quality linkage and not from a demonstration of robustness against a quality attribute, and it is bounded accordingly.

12 Conclusions

The step is characterized and ready for Stage 2. It delivers drug substance at 75 g/L with a product yield of 97.1% at set-point, closing the train at a cumulative product yield of 83.2%.

All 3 parameters are classified as key process parameters and none is linked to a critical quality attribute, so the step contributes no element to the design space, and it claims no proven acceptable range, no impurity clearance and no viral clearance.

The drug substance attributes measured after the step meet their acceptance criteria at commercial scale, with acidic charge variants (icIEF, AMV-3013) the tightest of the three at a Cpk of 4.26. Those estimates come from qualified scale-down models with every parameter drawn about its set-point at one sixth of its normal operating range, and they are confirmed at commercial scale during Stage 2.

Both deviations recorded during execution were investigated and retained, and neither changed a range, a classification or a conclusion. The outcome of this report rolls up into PCMR-001.

13 Deviations from the plan

Execution recorded 2 deviations against PCP-010. Both were investigated, both were impact assessed, and both were dispositioned as retained, with the register given in Table 13.1.

Table 13.1: Deviations recorded during execution of the study.

Deviation	Summary	Detected during	Disposition
DEV-010-01	Transmembrane-pressure excursion above NOR during concentration	in-process TMP monitoring	retained
DEV-010-02	Final DS concentration below target on one run; topped up	final concentration check	retained

Both deviations were found by the controls the plan put in place, one by the continuous in-process record of transmembrane pressure and the other by the concentration check at the end of the step. Neither was found after the fact, and neither required a retrospective assessment of material that had already moved on.

13.1 DEV-010-01 — Transmembrane pressure excursion above the normal operating range

Transmembrane pressure reached 14.6 psi during the first concentration phase, which is 0.6 psi above the upper edge of the normal operating range of 10–14 psi. The excursion was detected as it happened, on the pressure trend recorded by skid EQ-TFF-142.

The investigation reviewed the pressure and flux records for the run. Pressure rose through the first concentration phase and returned inside the control range once the diafiltration began, which is the signature of transient membrane fouling and not of a control failure, and the root cause was recorded on that basis. The skid was in calibration at execution, and no instrument fault was identified.

The impact assessment rests on two facts. The value reached is 0.4 psi inside the upper edge of the characterization range of 10–15 psi, so the excursion sits inside the region this study covers rather than outside it. Transmembrane pressure governs no quality attribute at this step, so there is no quality impact to assess. The run was retained, and no range or classification was changed.

13.2 DEV-010-02 — Final drug substance concentration below target

The second concentration phase finished at 71.5 g/L against a target of 75 g/L, a shortfall of 3.5 g/L. The value was inside the normal operating range of 70–80 g/L but below the target the batch record specifies, and it is 3.9 times the intermediate precision of the concentration method at the target, so it is not an artefact of the measurement.

The investigation traced the shortfall to residual hold-up volume. Buffer held in the recirculation loop and the membrane housing was recovered with the pool at the flush, and it diluted the pool below the concentration the phase had reached. The root cause is a property of the system rather than of the parameter, and it is the feature of the scale-down model discussed in §3.1.

The pool was returned to the skid, concentrated to target and confirmed by the ultraviolet absorbance method (AMV-3019). That correction is a normal in-process adjustment for this operation and needs no separate product disposition, because the concentration is measured directly on the product and the material stayed inside its normal operating range throughout. The run was retained, and the event is recorded here because a repeat would point at the flush volume and not at the parameter.

14 Appendix A — Analytical methods summary

Validated performance of the concentration method used at this step is given in Table 14.1. Performance of the size variant method (AMV-3011) and the charge variant method (AMV-3013) is held in their own validation reports and in the campaign method register, and is not reproduced here.

Table 14.1: Validated performance of the concentration method (AMV-3019).

Method	Title	Precision (%RSD)	LOQ	Accuracy (%)	Variance share
AMV-3019	Protein Concentration by A280 (UV)	1.2	0.5 g/L	99.6	0.22

15 Appendix B — Cumulative product yield across the train

The step by step product yield of the nominal batch is given in Table 15.1. The last row is this step, and the cumulative figure in that row is the overall product yield of the drug substance process.

Table 15.1: Step and cumulative product yield across the drug substance train.

Step	Unit operation	Step yield (%)	Cumulative yield (%)
3	Production Bioreactor	100.0	100.0
4	Harvest / Clarification	97.7	97.7
5	Protein A Chromatography	97.2	95.0
6	Low-pH Viral Inactivation	99.0	94.0
7	Cation Exchange Chromatography	95.2	89.5
8	Anion Exchange Chromatography	97.3	87.1
9	Small Virus Retentive Filtration	98.4	85.7
10	Ultrafiltration / Diafiltration (formulation)	97.1	83.2

16 Appendix C — Drug substance quality attribute register

The full register of drug substance critical quality attributes is given in Table 16.1, with the step that sets each attribute and the report that characterizes it. The register is reproduced here because the drug substance is expressed at this step and the release panel applies to the pool it delivers. No attribute in the register is set at this step.

Table 16.1: Drug substance quality attribute register, with the covering step and report.

CQA	Category	Critical-ity	Acceptance	Set by	Re- port
Afucosylation	glycosylation	H	2–13 % of glycans	Production Bioreactor	PCR-003
Galactosylation (total %Gal)	glycosylation	H	10–40 % (G1+G2 equiv.)	Production Bioreactor	PCR-003
High Mannose	glycosylation	VH	3–10 % of glycans	Production Bioreactor	PCR-003
Aggregates (HMW)	size	H	0–5 % HMW (SEC)	Production Bioreactor	PCR-003
Acidic charge variants (deamidation)	charge variant	VL	18–40 % (CEX/icIEF)	Production Bioreactor	PCR-003
Host Cell Protein (HCP)	process impurity	M-H	0–100 ng/mg	Production Bioreactor	PCR-003
Residual DNA	process impurity	VL	0–0.001 ng/dose	Production Bioreactor	PCR-003
Leached Protein A	process impurity	L-M	0–5 ppm	Protein A Chromatography	PCR-005
Viral clearance — XMuLV (enveloped)	viral safety	VH	16.7–30 log10 (cumulative)	Low-pH Viral Inactivation	PCR-006
Viral clearance — MVM (parvovirus)	viral safety	VH	8.6–20 log10 (cumulative)	Anion Exchange Chromatography	PCR-008

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